

EXP NO: 8

TESTING – TEST PLANS AND TEST CASES

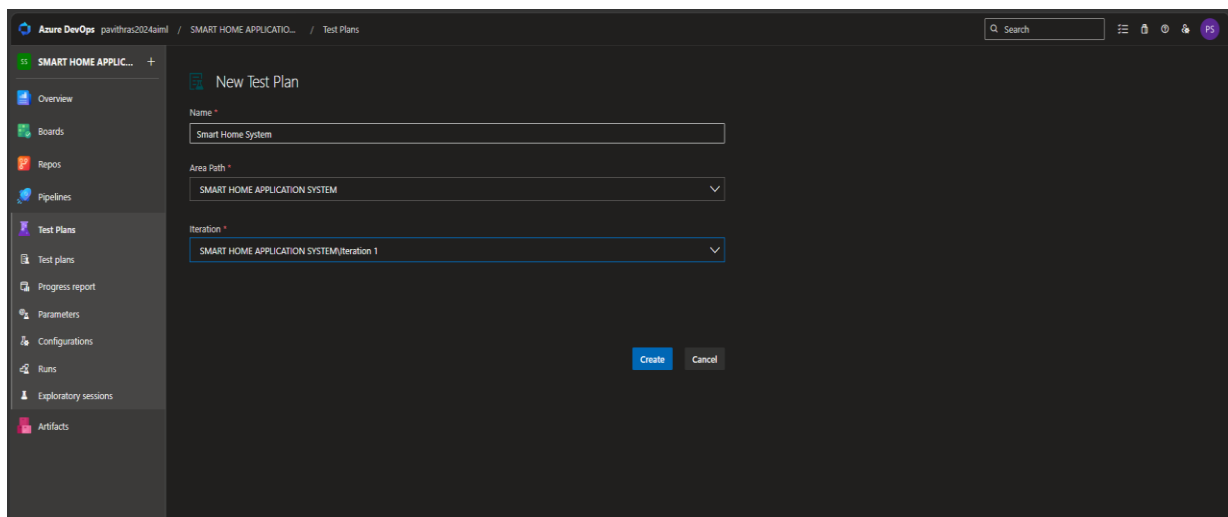
AIM:-

Test Plans and Test Case and write two test cases for at least five user stories showcasing the happy path and error scenarios in azure DevOps platform.

Test Planning and Test Case:

Test Case Design Procedure

1. New Test Plan



2. Test Suite

USER STORIES:-

- As a user, I want to sign up and log in securely so I can access and control smart home devices.
- As a user, I want to add and manage smart devices so I can monitor and control them remotely.
- As a user, I want to turn ON/OFF devices remotely for easy home automation.
- As a user, I want to schedule devices to operate automatically at specific times.
- As a user, I want to receive notifications and alerts from smart devices for security and monitoring.

Test Suite: TS01 – User Login & Registration

1. TC01 – Successful User Sign-Up

- Action:

- o Go to the Sign-Up page
- o Enter valid username, email, and password
- o Click "Sign Up"
- Expected Result:
- o User account is created successfully and redirected to dashboard
- Type: Happy Path

2. TC02 – Login with Valid Credentials

- Action:
- o Open Login page
- o Enter valid email and password
- o Click "Login"
- Expected Result:
- o User logged in successfully and dashboard displayed
- Type: Happy Path

3. TC03 – Sign-Up with Existing Email

- Action:
- o Open Sign-Up page
- o Enter already registered email
- o Click "Sign Up"
- Expected Result:
- o Error message: "Email already exists"
- Type: Error Path

4. TC04 – Login with Incorrect Password

- Action:
- o Enter invalid password
- o Click "Login"
- Expected Result:
- o Error message: "Invalid username or password"
- Type: Error Path

Test Suite: TS02 – Device Management

1. TC05 – Add Smart Device Successfully

· Action:

- o Login to system
- o Navigate to Device Management
- o Click "Add Device"
- o Enter device details
- o Click "Save"

· Expected Result:

- o Device added successfully and displayed in device list

· Type: Happy Path

2. TC06 – Remove Smart Device

· Action:

- o Select connected device
- o Click "Remove Device"

· Expected Result:

- o Device removed successfully from system

· Type: Happy Path

3. TC07 – Add Duplicate Device

· Action:

- o Try adding an already registered device

· Expected Result:

- o Error message: "Device already exists"

· Type: Error Path

4. TC08 – Add Device with Missing Details

· Action:

- o Open Add Device form
- o Leave mandatory fields empty
- o Click "Save"

· Expected Result:

- o Error message: "Required fields missing"

- Type: Error Path

Test Suite: TS03 – Device Control

1. TC09 – Turn ON Device Successfully

- Action:

- o Login to application

- o Select smart light

- o Click "ON"

- Expected Result:

- o Device status changes to ON

- Type: Happy Path

2. TC10 – Turn OFF Device Successfully

- Action:

- o Select active device

- o Click "OFF"

- Expected Result:

- o Device status changes to OFF

- Type: Happy Path

3. TC11 – Device Not Responding

- Action:

- o Try turning ON disconnected device

- Expected Result:

- o Error message: "Device unavailable"

- Type: Error Path

4. TC12 – Unauthorized Device Access

- Action:

- o Try controlling device without login

- Expected Result:

- o Access denied message displayed

- Type: Error Path

Test Suite: TS04 – Device Scheduling

1. TC13 – Schedule Device Successfully

· Action:

- o Login → Open Scheduling
- o Select device
- o Set operation time
- o Click "Save Schedule"

· Expected Result:

- o Schedule saved successfully

· Type: Happy Path

2. TC14 – Automatic Device Activation

· Action:

- o Wait until scheduled time

· Expected Result:

- o Device turns ON/OFF automatically

· Type: Happy Path

3. TC15 – Invalid Schedule Time

· Action:

- o Enter invalid time format
- o Click "Save"

· Expected Result:

- o Error message: "Invalid time entered"

· Type: Error Path

4. TC16 – Schedule Conflict

· Action:

- o Create overlapping schedules for same device

· Expected Result:

- o Error message: "Schedule conflict detected"

· Type: Error Path

Test Suite: TS05 – Notifications & Alerts

1. TC17 – Receive Security Alert

- Action:
 - o Trigger motion sensor alert
- Expected Result:
 - o Notification sent to user successfully
- Type: Happy Path

2. TC18 – Receive Device Status Notification

- Action:
 - o Turn ON/OFF device
- Expected Result:
 - o Device status notification received
- Type: Happy Path

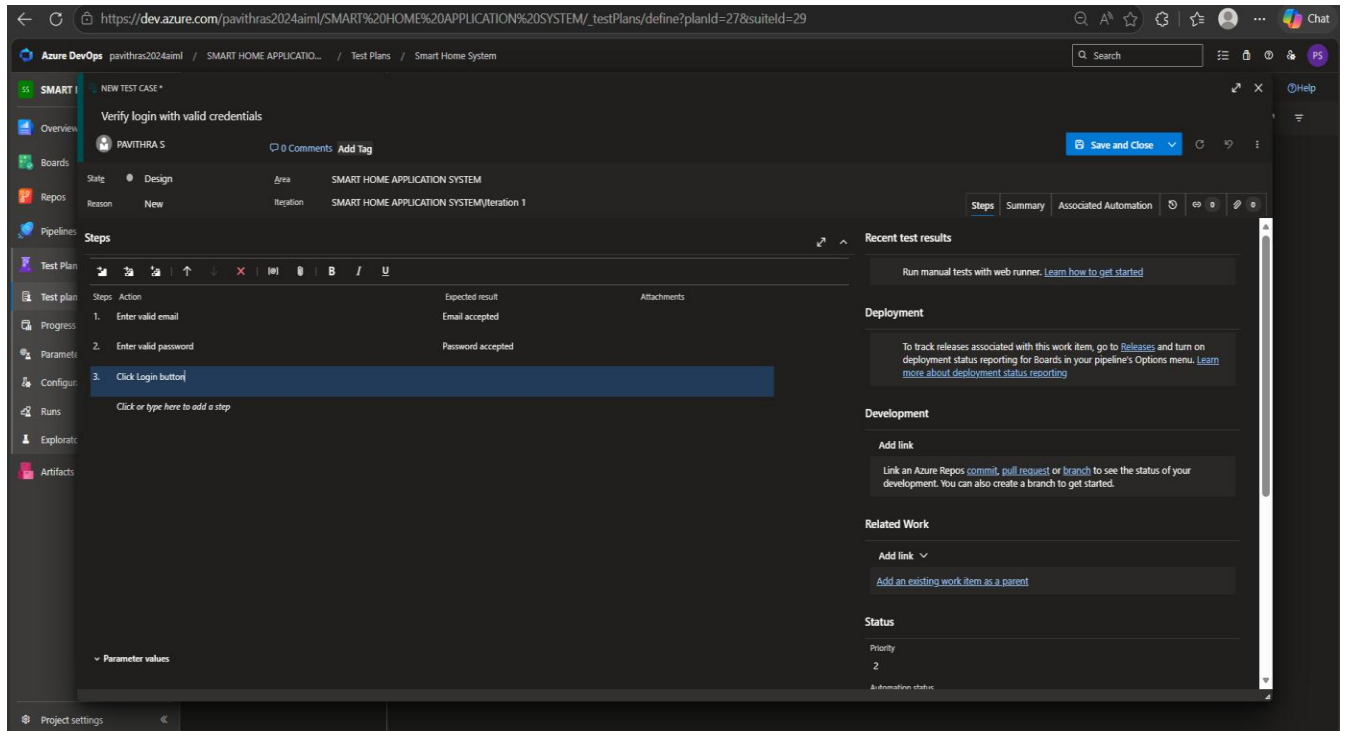
3. TC19 – Notification Delivery Failure

- Action:
 - o Disconnect internet during alert generation
- Expected Result:
 - o Error message: "Notification delivery failed"
- Type: Error Path

4. TC20 – Invalid Notification Settings

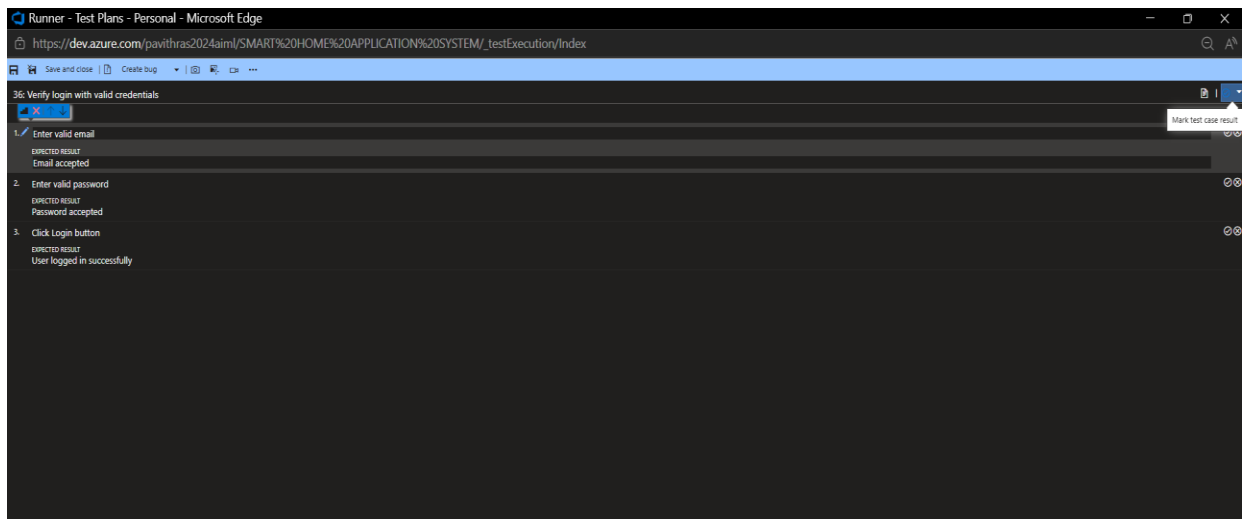
- Action:
 - o Save empty notification preferences
- Expected Result:
 - o Error message: "Invalid notification settings"
- Type: Error Path

3. Test Cases



4. Installation of tests

5. Running the test cases



6. Recording the test case

The screenshot shows the Azure DevOps Test Runs interface. The top navigation bar includes the Azure DevOps logo, the project name 'pavithras2024aiml', and the breadcrumb path 'SMART HOME APPLICATION / Test Plans / Runs / Test run 2 - Login Module Tests (Manual)'. A search bar and utility icons are on the right.

The left sidebar contains a navigation menu with options: Overview, Boards, Repos, Pipelines, Test Plans (selected), Test plans, Progress report, Parameters, Configurations, Runs, Exploratory sessions, and Artifacts. Below the menu is a 'Project settings' link.

The main content area is titled 'Test run 2 - Login Module Tests (Manual)' and shows a 'Completed' status. It includes a 'Run summary' section with a '100.00% pass rate' (1 Passed), manual test details (Test plan 27 - Smart Home System, Owned by PAVITHRA S), completion time (10/05/2026 13:10, 5m 54s), and release information. A 'Comment' section is also present.

Below the summary is a 'Test results' table with columns: Outcome, Test case, Run by, Priority, Configuration, Owner, Test suite, Iteration, Area, Duration, and Bugs. The table contains one row: 'Passed', '36 Verify login with valid crede...', 'PAVITHRA S', '2', 'Windows 10', 'PAVITHRA S', '29 Login Module T...', 'SYSTEM/iteration 1', 'LOCATION SYSTEM', '2m 29s', and '0'.

A notification box at the bottom right states: 'The new Test Runs experience is now generally available. Explore the new Test Runs interface with improved navigation, filtering, and analysis capabilities designed to streamline your testing workflow. Learn more Got it'.

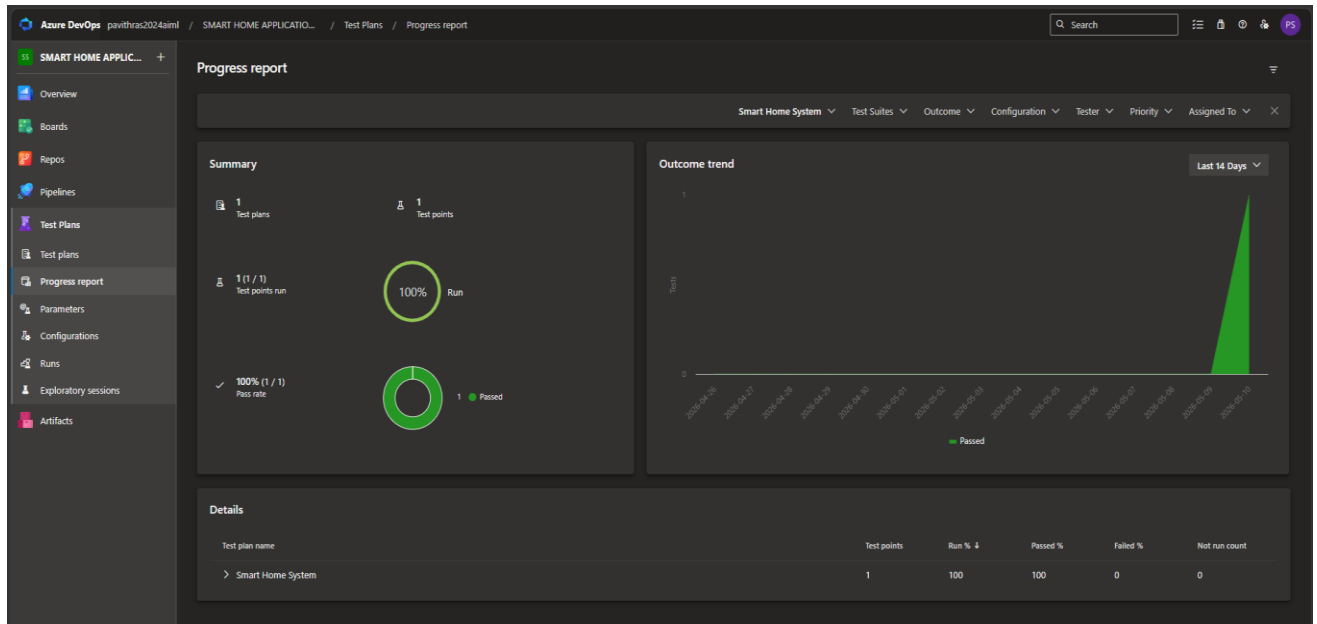
7. Creating the bug

The screenshot shows the 'NEW BUG' form in Azure DevOps. The title is 'Login button not responding'. It includes fields for 'No one selected', '0 Comments', and 'Add Tag'. A 'Save and Close' button is in the top right.

The form is divided into several sections:

- Repro Steps:** Test: Verify login with valid credentials, Priority: 2, Test file: None, Machine: None, Tested build: None, Error message: None, Stack trace: None. A note says: 'We support markdown, you can [convert this field](#).'
- Planning:** Resolved Reason, Story Points, Priority: 2, Severity: 3 - Medium, Activity, Effort (Hours) with Original Estimate, Remaining, and Completed fields.
- Deployment:** A note: 'To track releases associated with this work item, go to [Releases](#) and turn on deployment status reporting for Boards in your pipeline's Options menu. [Learn more about deployment status reporting](#).'
- Development:** Add link, Link an Azure Repos commit, pull request or branch to see the status of your development. You can also create a branch to get started.
- Related Work:** Add link, Add an existing work item as a parent.
- Tested By:** 36 Verify login with valid credentials, Updated 21m ago, Design.
- System Info:** System Info.

8. Test report summary



9. Progress report

10. Changing the test template

Result:

The test plans and test cases for the user stories is created in Azure DevOps with Happy Path and Error Path

